Holographic Principle and ER=EPR Conjecture

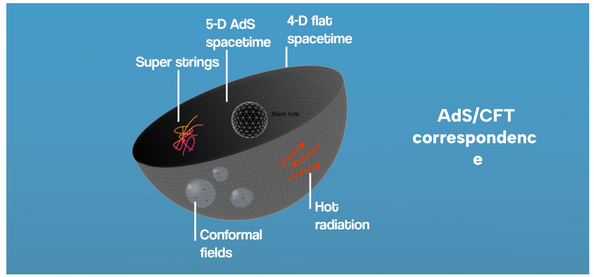
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DEPARTMENT OF PHYSICS

BACKGROUND

❖ **Holographic principle** : Total amount of information inside a volume of space cannot be larger than the amount of information that can be encoded on the boundary of that volume.



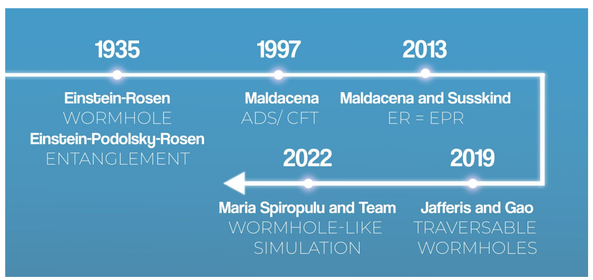
❖ **AdS/CFT correspondence** : Holographic duality between a bulk Anti-de Sitter space and a conformal field theory on its boundary.

❖ **Einstein-Rosen bridge** : A tunnel/bridge connecting two points in space time(also known as a wormhole).

❖ **Einstein-Podolsky-Rosen paradox** : Two entangled particles, once separated, measuring a property(spin) of one particle instantaneously determines the corresponding property of the other.

❖ **ER = EPR conjecture :** a wormhole (ER) is equivalent to a pair of maximally entangled particles (EPR).

TIMELINE



Maldacena's 1997 paper describing "AdS/CFT correspondence" has been cited by subsequent studies 22,000 times, more than twice a day on average

DATA

Bekenstein–Hawking entropy formula

$$S = \frac{kc^3}{4G\hbar}A$$

- S: Entropy of the black hole
- A: Area of the event horizon

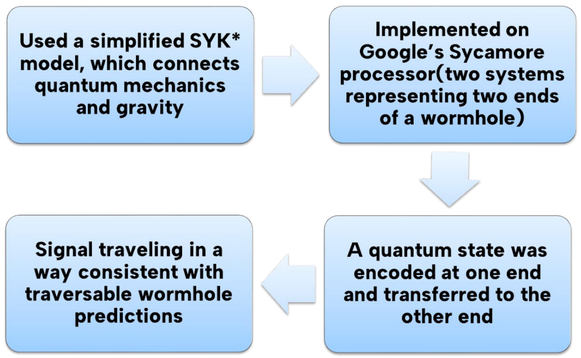
❖ This shows how **information scales with surface area**, not volume.

❖ The entropy also represents the amount of quantum entanglement between particles (or regions of space-time).

❖ In ER=EPR, this **entanglement** (EPR pairs) creates the **geometry of space-time** bridges (wormholes).

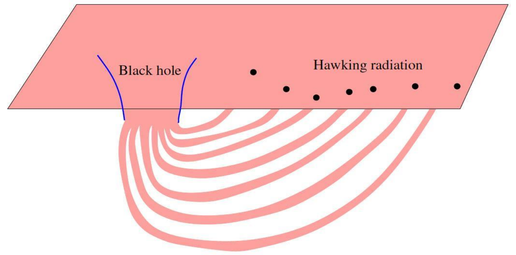
2022 Wormhole simulation

❖ In 2022, researchers simulated aspects of a wormhole's behavior on Google's Sycamore quantum computer



❖ This implies that **wormholes** are a geometric representation of the **entanglement structure** encoded in the holographic boundary.

* SYK model describes a system of many particles interacting randomly with each other(Highly disordered and all-to-all interactions)



❖ Hawking radiation may be connected to microscopic wormholes (ER bridges) through entanglement, suggesting a mechanism for information transfer encoded in quantum correlations

DISCUSSION

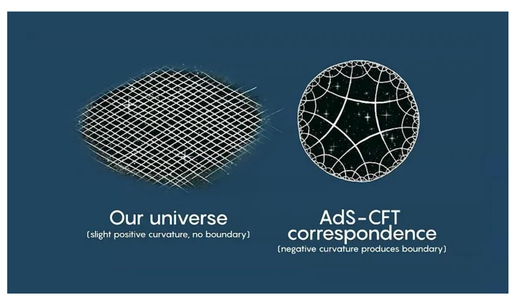
"I think it's about time we got out from under the protective layer of AdS space and open up into the world that might have more to do with cosmology, De Sitter space is another beast."

-Leonard Susskind

1. Locality means the impossibility of sending signals faster than the speed of light. Neither wormholes nor EPR correlations give rise to real violations of locality.

2. These concepts suggest a **reformulation of locality**, where space-time and entanglement are interwoven, allowing quantum correlations to manifest without violating causality or classical locality.

3. De Sitter space is associated with an expanding universe and lacks the mathematical structure that AdS space provides, complicating holographic formulations.



CONCLUSIONS

- The Holographic Principle and ER=EPR conjecture suggest a profound connection between quantum mechanics, space-time, and gravity, offering a pathway to understanding quantum gravity.
- These concepts provide potential resolutions to the black hole information paradox by linking entanglement, wormholes, and the holographic encoding of information.
- Recent advancements, such as lab analogs and quantum simulations, bring theoretical ideas closer to experimental verification, bridging the gap between theory and observation.
- What we perceive as a three-dimensional space could fundamentally be encoded in a two-dimensional form at the boundary of our universe.

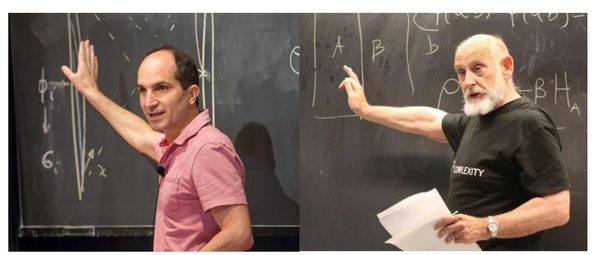
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Juan Maldacena (left) and Leonard Susskind (right)

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